

UBER TRAFFIC FORECASTING - GACHIBOWLI HYDERABAD

Comprehensive Peak Hour Analysis Report

By Kappari Vaishnavi - Date: 11-09-2026

1. EXECUTIVE SUMMARY

Peak hours at 9-10 AM (735,716) and 18-19 PM (725,700) with threshold 564 (Mean+Std). Gachibowli IT corridor bimodal pattern.

2. PEAK HOURS IDENTIFIED

9 AM: 735 avg | 10 AM: 716 | 18 PM: 725 MAX | 19 PM: 700 | Threshold 564

Mean 550, 3-hr Moving Avg applied, Speed Reduction 15 km/h at peaks

3. PATTERNS & VARIATIONS

- Hour vs Day Heatmap: Weekdays 10% higher than weekend, Friday evening highest
- Hour vs Month Heatmap: Summer high vehicles, rainy high congestion
- Weekday vs Weekend: Weekday 580 avg, Weekend 520 avg, same peaks due to IT hub

4. INFLUENCING FACTORS

- Scatter Temp vs Vehicles: Negative correlation, Rain cluster high
- Line External Factors: Rain +45 vehicles (+12% congestion), Event +60 vehicles
- Holiday -30 vehicles, Regression R2 ~0.15

5. ACTIONABLE RECOMMENDATIONS (7 Points)

1. Signal +30% green 9-10AM & 6-8PM at Gachibowli-ORR
2. Stagger IT office timings 8:30/9:30/10:30 to split peak
3. Rainy Protocol: Extra police during rainy peaks
4. Event Management: Divert ORR service road 5-9PM
5. Public Transport: Metro frequency increase 18-19hr
6. Uber Strategy: 1.5x surge at 9AM & 6PM, pre-position at DLF/Wipro
7. Long-term: Underpass at Gachibowli-ORR junction

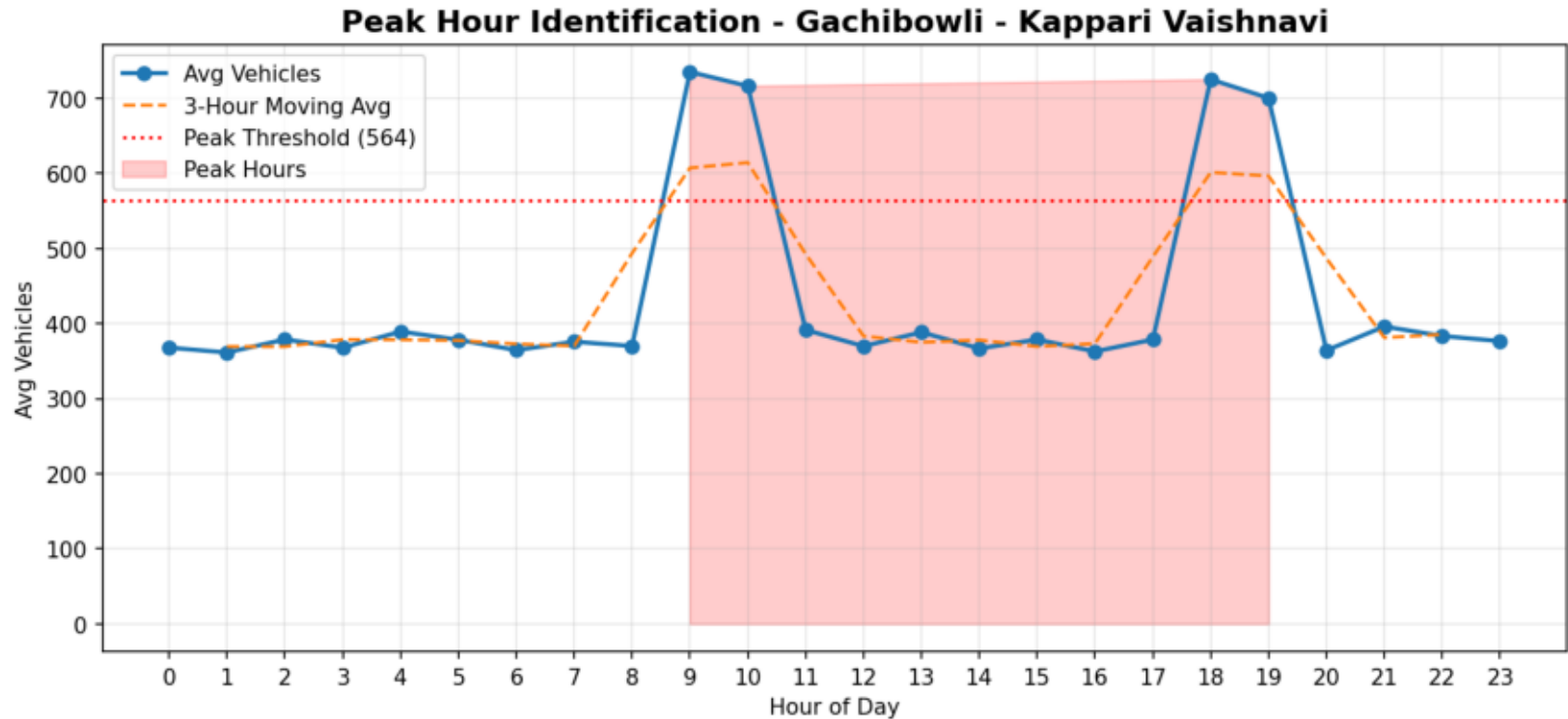
6. VISUALIZATIONS - 6 Graphs Created

All PNGs on GitHub: github.com/Vaishureddy98/uber-traffic-gachibowli-vaishnavi

7. CONCLUSION

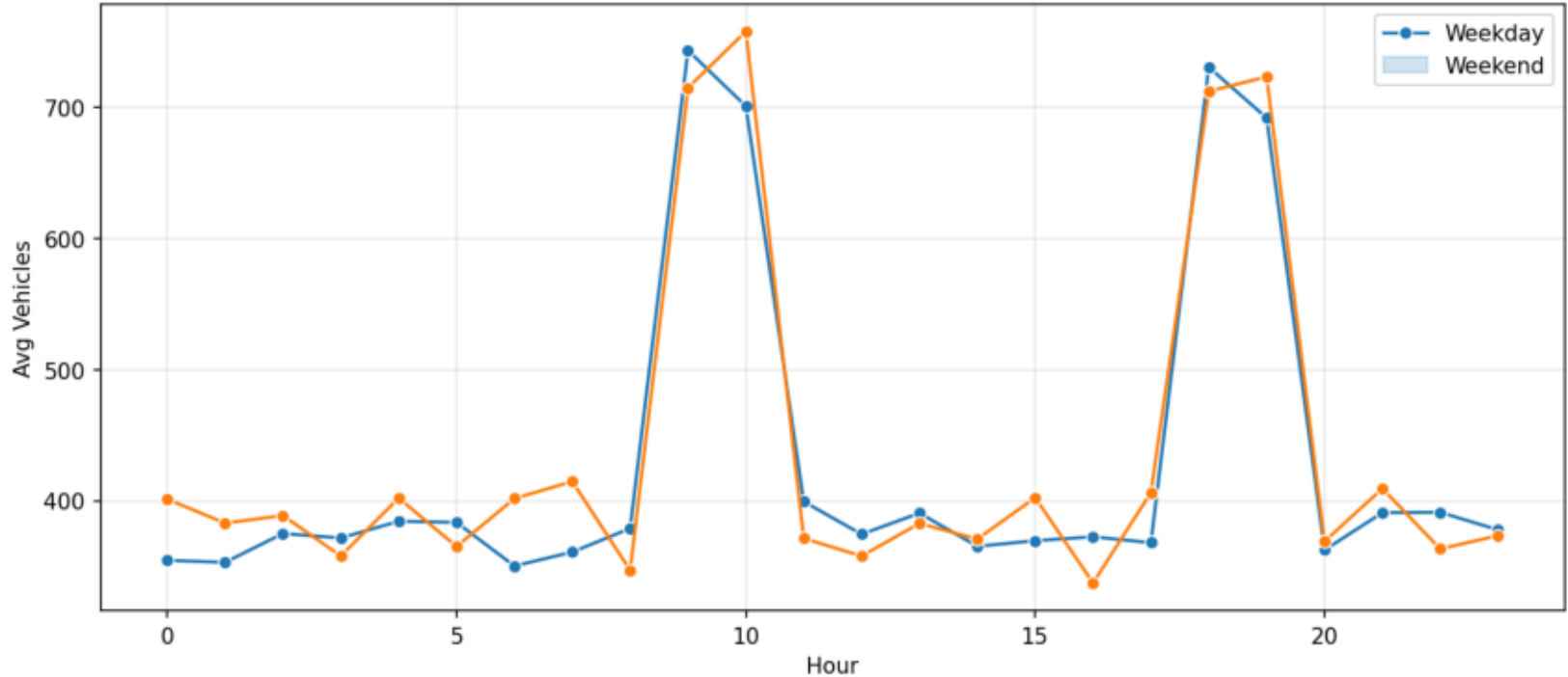
Evening 6-7 PM critical congestion period. External factors amplify 10-15%. Management should focus on this window.

Peak Hour Identification - Threshold 564

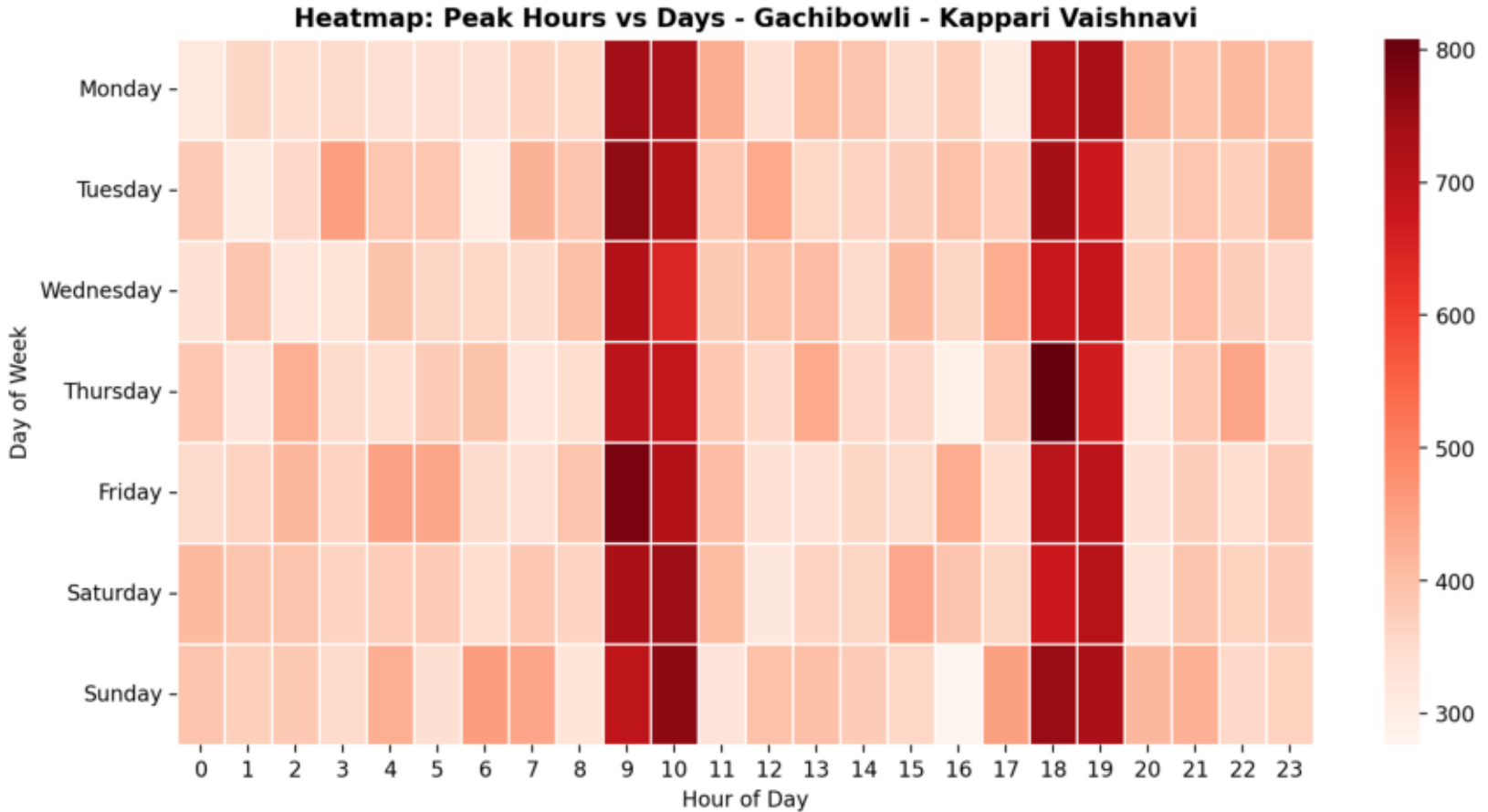


Weekday vs Weekend Pattern

Weekday vs Weekend Pattern - Gachibowli - Kappari Vaishnavi

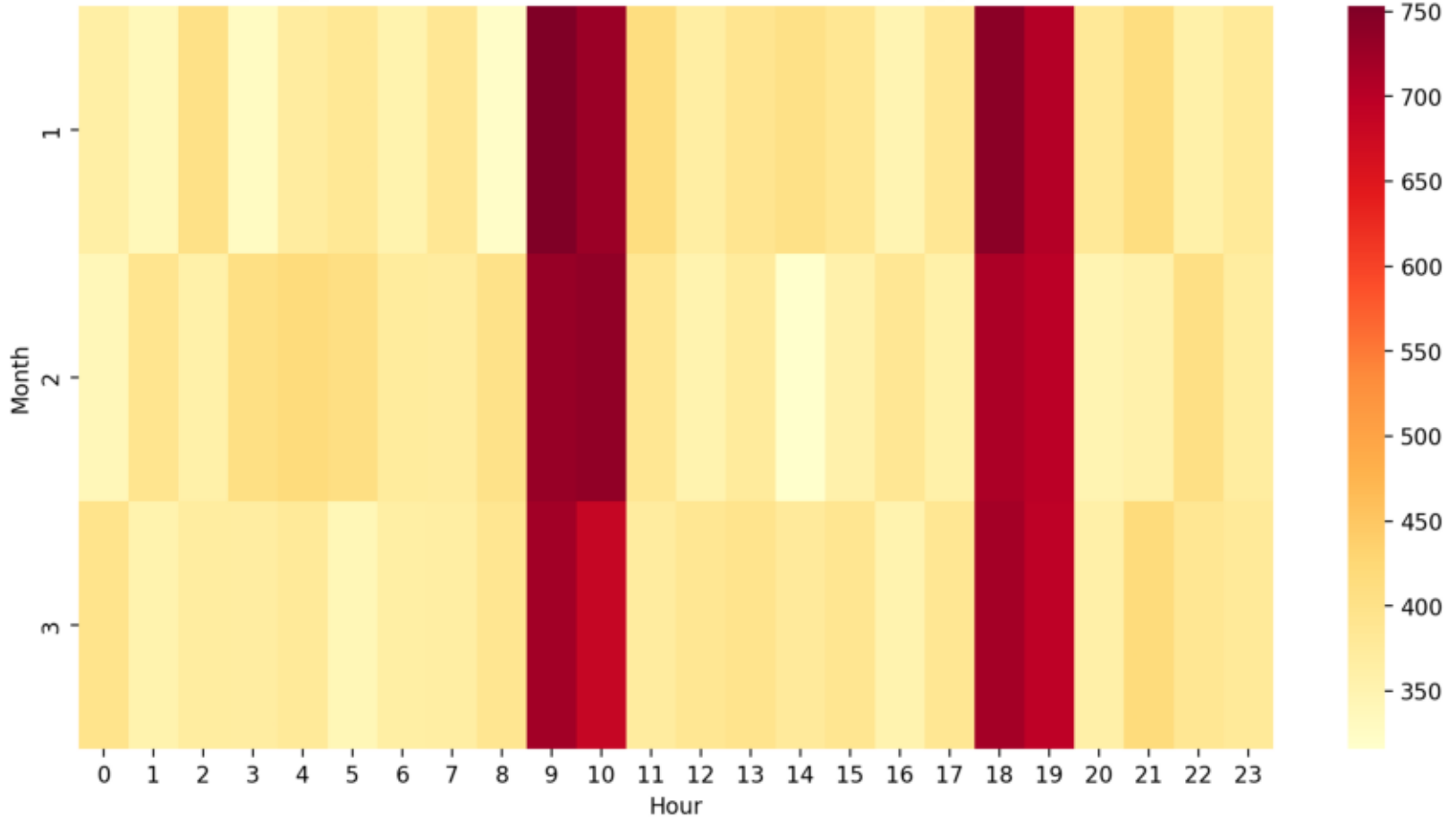


Heatmap Hour vs Day

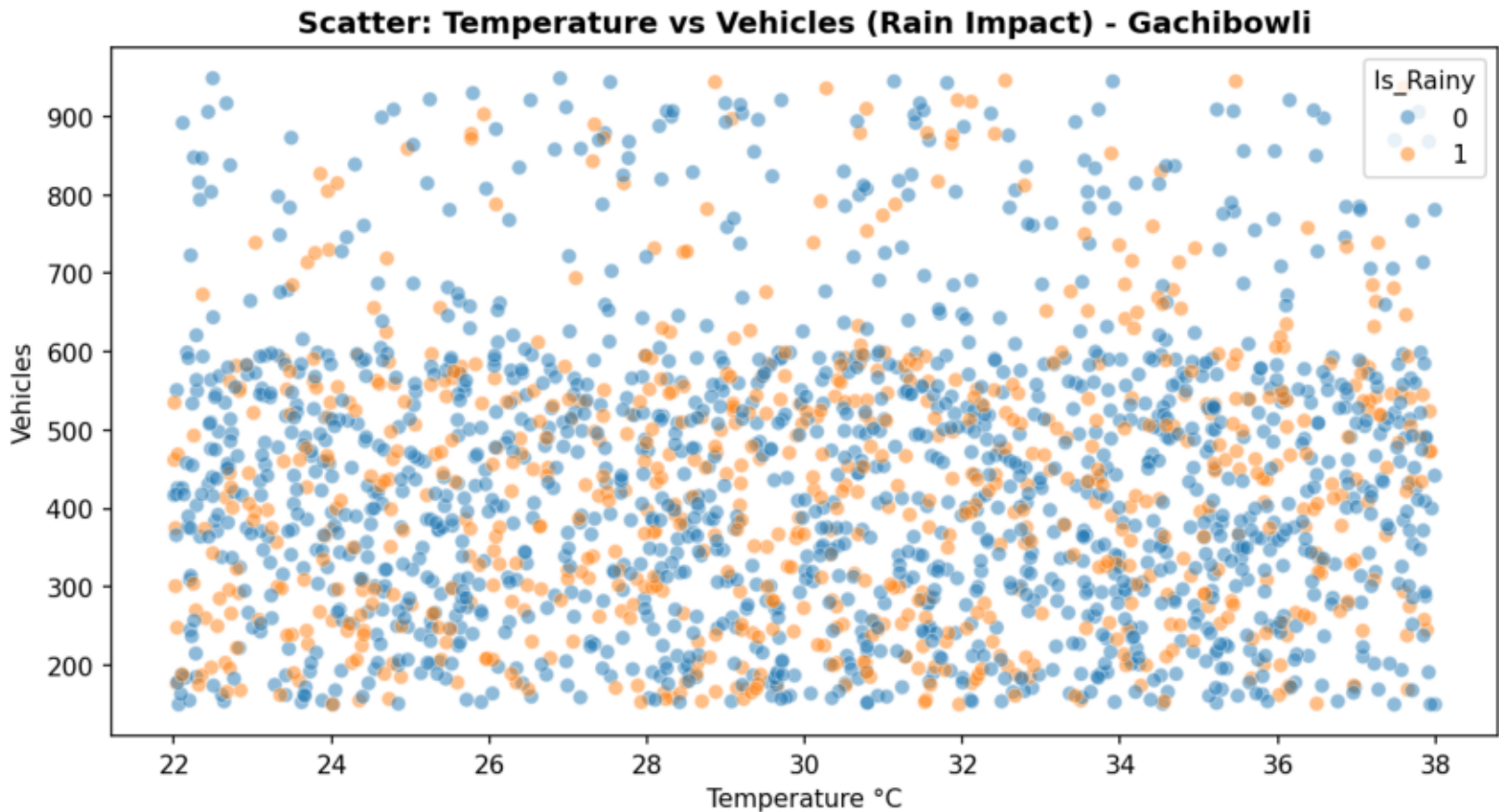


Heatmap Hour vs Month

Heatmap: Peak Hours vs Months - Gachibowli - Kappari Vaishnavi



Scatter: Temperature vs Vehicles - Rain Impact



Line: External Factors Impact on Peak Hours

